Head & Neck Osteology: Mandible & Associated Structures (Set 2)

1. The mandible is the strongest and lowest bone of the face. Its dynamic horizontal horseshoe-shaped component that carries the lower teeth is the:
a. Ramus
b. Angle
c. Condyle
d. Body
e. Coronoid process
2. On the anterior surface of the mandible, the two halves of the bone meet and fuse in the midline during the first year of life to form the:
a. Mental foramen
b. Alveolar border
c. Symphysis menti
d. Oblique line
e. Mylohyoid line
3. The triangular elevation located at the lower end of the symphysis menti, which forms the physical prominent base of the chin, is the:
a. Mental tubercle
b. Mental protuberance
c. Mandibular notch
d. Lingula
e. Mylohyoid groove
4. The mental foramen is a key local anesthesia landmark on the outer surface of the mandibular body. In adults, it is typically located below the interval between which teeth?
a. Central and lateral incisors
b. Canine and first premolar
c. First and second premolars
d. Second premolar and first molar
e. First and second molars
5. The mental foramen transmits the mental nerve and vessels to the lower lip and chin skin. The mental nerve is a direct branch of the:
a. Incisive nerve
b. Buccal nerve
c. Lingual nerve
d. Facial nerve
e. Inferior alveolar nerve
6. Running backward and upward from the mental tubercle is a faint bony ridge that continues into the anterior border of the ramus. This is the:
a. Mylohyoid line
b. Oblique line
c. Incisive fossa
d. Digastric fossa
e. Temporal crest

7. On the inner (posterior) surface of the mandibular body near the midline, there are two pairs of small bony projections that serve as muscle attachments. These are the:
a. Mental tubercles
b. Genial tubercles (Mental spines)
c. Mandibular angles
d. Condylar processes
e. Pterygoid foveae
8. The superior pair of genial tubercles gives origin to which extrinsic tongue muscle, responsible for protruding the tongue?
a. Hyoglossus
b. Styloglossus
c. Palatoglossus
d. Genioglossus
e. Geniohyoid
9. The inferior pair of genial tubercles gives origin to the geniohyoid muscle, which is functionally classified as a:
a. Suprahyoid muscle
b. Infrahyoid muscle
c. Muscle of mastication
d. Muscle of facial expression
e. Palatal muscle
10. Situated on the inner surface of the mandibular body is a sharp, oblique bony ridge that runs backward and upward toward the third molar tooth. This is the:
a. Oblique line
b. Temporal crest
c. Mylohyoid line
d. Buccinator crest
e. Mandibular notch border
11. The mylohyoid line gives origin to the mylohyoid muscle, which forms a vital soft tissue barrier in dentistry known as the:
a. Roof of the mouth
b. Floor of the mouth
c. Tonsillar pillar
d. Buccal vestibule wall
e. Sublingual caruncle base
12. The mylohyoid line divides the inner surface of the mandible into two distinct depressions. The shallow depression located superior to the line anteriorly is the:
a. Submandibular fossa
b. Sublingual fossa
c. Digastric fossa
d. Pterygoid fovea
e. Incisive fossa
13. The sublingual fossa houses the sublingual salivary gland. The deeper depression located inferior to the mylohyoid line posteriorly is the:
a. Sublingual fossa
b. Submandibular fossa
c. Digastric fossa
d. Retromolar fossa

e. Mandibular notch

14. The submandibular fossa houses the large submandibular salivary gland. Located on the inner surface near the lower border of the symphysis menti is a small, oval depression called the:

a. Sublingual fossa
b. Submandibular fossa
c. Pterygoid fovea
d. Incisive fossa
e. Digastric fossa

15. The digastric fossa gives attachment to the anterior belly of the digastric muscle, which acts mechanically to:

a. Elevate the mandible
b. Protrude the mandible
c. Retract the mandible
d. Depress the mandible (open the mouth)
e. Move the mandible laterally

16. The vertical, quadrilateral plate of bone projecting upward from the posterior part of the mandibular body is the:

a. Ramus
b. Angle
c. Condyle
d. Coronoid process
e. Alveolar process

17. The inner surface of the mandibular ramus features a prominent opening that leads into a canal running through the bone. This opening is the:

a. Mental foramen
b. Mandibular foramen
c. Infraorbital foramen
d. Foramen ovale
e. Incisive foramen

18. Guarding the anterior margin of the mandibular foramen is a sharp, tongue-shaped bony projection that serves as an attachment for the sphenomandibular ligament. This is the:

a. Mylohyoid line
b. Oblique line
c. Lingula
d. Condyle
e. Coronoid process

19. To perform an inferior alveolar nerve block (dental anesthesia), the clinician deposits anesthetic solution near the mandibular foramen. This solution must target the nerve before it enters the:

a. Mandibular canal
b. Mental canal
c. Incisive canal
d. Mylohyoid groove
e. Pterygomandibular space exit

20. Running downward and forward from the mandibular foramen is a distinct narrow groove that contains a nerve supplying the mylohyoid muscle. This is the:

a. Mandibular groove
b. Oblique groove
c. Mental groove
d. Mylohyoid groove
e. Pterygoid groove

21. The upper end of the mandibular ramus features two prominent processes separated by a deep, curved depression. This deep depression is the:

a. Coronoid notch
b. Mandibular notch (Incisura)
c. Mylohyoid groove
d. Mental foramen
e. Mandibular canal

22. The anterior process at the top of the ramus is a sharp, triangular projection that receives the insertion of the powerful temporalis muscle. This is the:

a. Condylar process
b. Styloid process
c. Mastoid process
d. Coronoid process
e. Lingula

23. The posterior process at the top of the ramus is expanded into an articular head that forms the temporomandibular joint (TMJ). This process is the:

a. Coronoid process
b. Styloid process
c. Condylar process (Head of mandible)
d. Angle of mandible
e. Mental spine

24. Immediately inferior to the smooth articular surface of the condyle is a constricted zone known as the neck of the mandible. On its anterior aspect is a small depression called the:

a. Digastric fossa
b. Submandibular fossa
c. Pterygoid fovea
d. Mandibular notch
e. Sublingual fossa

25. The pterygoid fovea receives the primary insertion of which muscle of mastication, responsible for acting as the primary opener of the jaw?

a. Masseter muscle
b. Medial pterygoid muscle
c. Lateral pterygoid muscle
d. Temporalis muscle
e. Buccinator muscle

26. The lateral surface of the mandibular ramus is flat and marked by faint oblique ridges. It provides insertion for which powerful muscle of mastication?

a. Temporalis
b. Masseter
c. Medial pterygoid
d. Lateral pterygoid
e. Mylohyoid

27. The medial (inner) surface of the ramus near the angle of the mandible features a rough, ridged area that provides insertion for which muscle of mastication?
a. Lateral pterygoid muscle
b. Medial pterygoid muscle
c. Masseter muscle
d. Temporalis muscle
e. Digastric muscle
28. The junction where the lower border of the mandibular body meets the posterior border of the ramus is a prominent practical landmark known as the:
a. Inion
b. Bregma
c. Pterion
d. Asterion
e. Angle of the mandible (Gonion)
29. In a young child, the angle of the mandible is characteristically obtuse (around 140 degrees). In a normal, dentulous adult, the angle changes to become approximately:
a. 90 degrees
b. 110 to 120 degrees
c. 150 degrees
d. 170 degrees
e. 180 degrees
30. In old age, following the loss of teeth and subsequent absorption of the alveolar process, the mandibular angle becomes:
a. Acute (90 degrees)
b. Right-angled
c. Obtuse again (around 140 degrees)
d. Completely flattened
e. Segmented
31. The upper border of the mandibular body contains sixteen deep sockets to hold the roots of the lower teeth. This specialized border is the:
a. Basal border
b. Alveolar process (border)
c. Coronoid notch
d. Mental ridge
e. Internal oblique line
32. Following tooth extraction, the lack of mechanical stress causes a specific physiological alteration in the bone structure of the mandible. This process is:
a. Hypertrophy of the ramus
b. Atrophy and resorption of the alveolar process
c. Condylar elongation
d. Ossification of the mandibular canal
e. Hyperplasia of the mental spines
33. The facial artery can be easily palpated on the mandible to check a patient's pulse. It crosses the lower border of the bone immediately anterior to which muscle?
a. Temporalis
b. Lateral pterygoid
c. Buccinator
d. Masseter

e. Mylohyoid

34. Fractures of the mandible can occur at various points. Due to the transmission of mechanical forces across the arch, an impact to one side of the jaw frequently causes a fracture of the neck of the condyle on the:

a. Identical same side
b. Completely opposite side (contralateral side)
c. Symphysis menti directly
d. Coronoid process on the same side
e. Mental foramen region only

35. The weakest parts of the mandible where fractures occur most commonly are the neck of the condyle, the angle, and the region of the body containing the socket of the:

a. Central incisor
b. Lateral incisor
c. Canine tooth
d. First premolar
e. First molar

36. A fracture passing through the mandibular canal can cause direct trauma to the inferior alveolar nerve. This manifests clinically as a loss of sensation (numbness) affecting the:

a. Posterior one-third of the tongue
b. Skin of the forehead
c. Upper lip and maxilla
d. Lower lip, lower teeth, and skin of the chin
e. Hard palate mucosa

37. Dislocation of the temporomandibular joint occurs when the condyle slips completely out of the mandibular fossa. The condyle almost always slips in which direction?

a. Posteriorly into the external acoustic meatus
b. Superiorly into the middle cranial fossa
c. Anteriorly over the articular eminence into the infratemporal fossa
d. Medially into the pterygopalatine fossa
e. Laterally through the skin

38. When the mandible is dislocated anteriorly, the patient presents clinically with an inability to:

a. Open the mouth
b. Protrude the jaw
c. Close the mouth (the jaw remains locked open)
d. Retract the tongue
e. Swallow fluids

39. The lower border of the mandible is thick and rounded, forming the base of the jaw. It is composed of dense compact bone that resists fractures, making it a reliable landmark for:

a. Local anesthetic injections
b. Extraoral surgical approaches and plating
c. Placement of orthodontic bands
d. Tracking the path of the lingual nerve
e. Palpating the parotid duct

40. The small triangular area of bone located immediately posterior to the lower third molar tooth, bounded by the buccinator and temporal crests, is the:
a. Sublingual fossa
b. Retromolar fossa (triangle)
c. Pterygoid fovea
d. Mandibular notch
e. Digastric basin

Answer Key Table (Set 2 - Sequence 2 Loop)

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40
d	a	b	c	d
c	c	c	b	b
a	b	a	b	c
b	b	d	a	c
b	d	d	a	c
a	a	c	d	b
c	b	d	b	c
d	a	c	c	e